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/ [Test 4 \(topics 10-12: Applications of Lagrange's theorem,Coding theory,Coding theory and group codes\)](#)

Started on	Friday, 29 November 2024, 2:15 PM
State	Finished
Completed on	Friday, 29 November 2024, 2:28 PM
Time taken	13 mins 27 secs
Marks	3.00/3.00
Grade	3.00 out of 3.00 (100%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Correct

Mark 1.00 out of 1.00

If p and q be prime numbers, let $n = p \cdot q$, and write (n, x) and (n, y) for the public and private key in an RSA key pair associated to n .

In the list below we have written suggestions for $(x, y, p, q, p \cdot q)$. Some of them give valid RSA keys, some of them do not for different reasons.

Select all options that give us a valid key pair. Leave the invalid options un-checked.

- ☒ $x = 20271383, y = 22663367, p = 3557, q = 7741, n = 27534737$
- ☐ $x = 7057729, y = 1240783, p = 6007, q = 3271, n = 19648897$
- ☐ $x = 42185665, y = 31377985, p = 7249, q = 6329, n = 45878921$
- ☐ $x = 35131943, y = 15273131, p = 7579, q = 7583, n = 57471557$
- ☒ $x = 17570399, y = 30362099, p = 7451, q = 6451, n = 48066401$
- ☐ $x = 12903033, y = 3215451, p = 6113, q = 2437, n = 14897381$
- ☐ None of the above

You are free to use this calculator for products $\mod (n)$. For example, if you want to compute $(7634 \cdot 99833 \% 8883993)$, you should set **Factor 1**=7634, **Factor 2**=99832, and **Modulus**=8883993.

Factor 1:
factor 2:
Modulus:
Result:

You are free to use this calculator for computing powers $\mod (n)$. For example, if you want to compute $(7634^{99382} \% 8883993)$, you should set **Base**=7634, **Exponent**=99382, and **Modulus**=8883993.

Base:
Exponent:
Modulus:
Result:

Question **2**

Correct

Mark 1.00 out of 1.00

Given a noisy binary symmetric channel. The probability of making an error when transmitting a single bit is $p = 0.074$.

The bitstring $w = 1010000$ is transmitted.

In all of the answers below, all answers must be rounded off to the three decimals.

a) What is the probability of making no errors transmitting w ?

Your last answer was interpreted as follows:

0.584

b) What is the probability of making at most 2 bit errors transmitting w ?

Your last answer was interpreted as follows:

0.989

b) What is the probability of transmitting 1010000 and receiving $r = 0010001$?

Your last answer was interpreted as follows:

0.0037

Question 3

Correct

Mark 1.00 out of 1.00

We use the generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}$$

to define the encoding function

$$E : (\mathbb{Z}/2)^{\times 4} \rightarrow (\mathbb{Z}/2)^{\times 8}$$

by the formula

$$E(w) = w \cdot G.$$

An encoded message $c = E(w)$ is transmitted over a noisy channel and received as

$$r = 01100111.$$

a) Compute the syndrome of $r =$

Your last answer was interpreted as follows:

0100

b) Compute the error pattern e , so that $r = c + e$.

$e =$

Your last answer was interpreted as follows:

00000100

c) Compute the encoded message $c =$

Your last answer was interpreted as follows:

01100011

Syntax guide: Give your answers as bitstrings E.g. if you think the answer to a question is "1011", then you should enter **1011**.

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 413

Signing code:

Your last answer was interpreted as follows:

26389

[◀ Test 3 \(topics 7-9: Recurrence relations, Introduction to group theory, The multiplicative group of integers mod\(n\) and Lagrange's theorem\)](#)

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